

REVISED 2026-09-17 — corrected sample selection

This checkpoint has been **re-run and replaced in place**. Two defects in the sample selection were found after the first issue (see `beam_metric_outliers_checkpoint`) and have been fixed here. Every number below is from the corrected run.

1. Calibration data was being fit as if it were beam data. 23 of the 226 files in this window are calibration files — the receiver was on a VNA, noise source or load, not on the antenna. Nothing in the mask chain was a campaign gate, so those samples entered the fit. They carried a **median 71.3% of the residual power** across the 101 channels.

The exclusion is built from the **per-sample** `metadata/rfswitch` **stream**, not from the `rfswitch_dominant` column of `curation/file_state.csv` . That column is a file-level majority vote, and 14 of the 23 calibration files are majority `RFANT` , so a dominant-state test keeps them (`corr_20260717_191940Z.h5` is `{RFANT: 124, RFNON: 111, RFAMB: 3}`). Going per sample also keeps 1938 genuine on-antenna samples that a whole-file exclusion would have thrown away.

2. EL_SOLUTION_GLITCH . `pointing_table@v1.2+45f8059` adds a bit marking samples where the IMU elevation solution jumped faster than the drive can move. Small in aggregate (~0.002 on the median) but locally decisive.

Net effect on the mask: **25352 → 22969 samples** (−2200 off-antenna, −183 glitch), and on the headline:

before after	--- --- ---	median normalized RMS (corrected geometry)	0.9029	0.4691	
channels below 0.5	2	61	median normalized RMS (old geometry)	0.9115	0.5502

3. Section 6 is un-blocked and rewritten. Q8 is closed; `el` is the boresight zenith angle. The `|el| ≈ 180` "wrap cluster" is boresight-on-transmitter, not an angle-wrap artifact, and is now counted.

`fit_beam_v2.py` , which generated the reports read here, lived only in `/tmp` and was unversioned until this revision. It now sits next to this notebook.

D2 review checkpoint: beam-fits-v2 -- geometry stopgap fix

Milestone: D2, measured $A(\nu, \theta, \phi)$ + uncertainties over stated coverage, consuming `pointing_table@v1`.

What changed since the first (bad) run: the first `beam-fits-v2` run held the TX geometry (heading/alpha) fixed at `v007`'s old consensus fit, which was calibrated against **raw motor pointing**, while feeding it `pointing_table@v1` -**corrected** az/el for the templates. That is a demonstrable frame mismatch (measured directly: a clean +34.8 deg elevation offset, std 0.27 deg, plus a non-constant -30.6 deg mean / 19.8 deg std azimuth offset from the documented slip correction) -- not a beam-quality result. This notebook re-fits the geometry against `pointing_table@v1` itself (orthogonal to geometer's separate joint transmitter+antenna position MCMC), then re-runs the beam fit against the corrected geometry.

Window: files[-227:-1] of the campaign glob, 226 files, `corr_20260717_185100Z.h5` .. `corr_20260718_032126Z.h5` (doc cites 227 for the same era at `marjum-2026-07/INDEX.md:962` -- one-file boundary-inclusivity rounding, not a different window). 64/226 dropped as comb-off; 162 files contribute samples.

TX-identity caveat (unresolved, carried forward unchanged): this fit assumes the corrected identity -- the 1.953125 MHz, 8-channel-locked comb is the transmitter (rfi-analyst's rotation-modulation test). Not settled either way by this analysis. If the identity resolves the other way, this is a near-field self-comb map of an instrument-internal radiator, not a far-field transmitter beam map -- same numbers, opposite physical meaning.

Receiver-regime caveat (unresolved, carried forward unchanged): rf-calibrator's B7 finding places a receiver regime change (T_{rx} 212->539 K) between 07-17 19:42 and 07-18 01:24 UTC, bracketing nearly this entire window. The channel-differenced `measured_tx` quantity is largely gain-ratio-invariant to first order, but this has not been separately verified -- a material uncertainty on absolute amplitude scale, not on beam shape.

Metric convention (Aaron, 2026-09-15): every data-vs-model comparison in this notebook is judged on **normalized RMS** -- residual RMS divided by the RMS of the data being fitted, after a single free amplitude. Correlations, rank correlations, peak azimuths and circular means appear only as diagnostics to explain **why** a fit behaves as it does; they are never the comparison itself. Two earlier conclusions in this notebook were wrong precisely because a diagnostic statistic was used as the metric (section 13's Pearson correlation, section 15's circular mean); both are marked withdrawn where they appear.

JAX is not using 64-bit precision. This will dramatically affect numerical precision at even moderate L.

JAX is not using 64-bit precision. Results will diverge from healpy even at moderate nside and can overflow for nside > 8192. See the README on numerical precision.

Loaded all four inputs OK.

1. Geometry: old (raw motor pointing) vs new (`pointing_table@v1`)

	old	new (v1 pointing)	delta
heading_x	0.147916	0.905382	0.757465
heading_y	0.144084	-0.418135	-0.562218
heading_z	-0.978448	-0.073808	0.904640
alpha_deg	45.090557	10.073214	-35.017343
joint_normalized_rms	0.414542	0.324718	-0.089824
n_channels_kept	73.000000	73.000000	0.000000
n_channels_rejected	0.000000	0.000000	0.000000

1a. The surveyed geometry -- and a retraction of the section above

The transmitter is essentially directly below the antenna. Taking both positions from the archive in the same ENU frame (`transmitter_position@v2` and `horizon_profiles` ' `antenna_enu_m` , 91 m era), the transmitter is **93.0 m below** the antenna at **6.9 m** horizontal range: a look angle of **85.8 deg below horizontal**, near-nadir.

The machinery can represent this -- `ground_heading` builds `[E, N, -height]` and `simulate_hfss_coupling` rotates that fixed heading into the antenna frame, so nothing assumes "el = 0 points at the transmitter." But the heading actually in use does not describe the real experiment, and **the refit performed in section 1 made it much worse.**

```
antenna ENU [1655.868 2030.923 1777.397]
transmitter ENU [1652.198 2025.104 1684.384]
antenna -> TX: dE -3.67 dN -5.82 dU -93.01 m
horizontal range 6.88 m, vertical drop 93.01 m
look angle 85.8 deg below horizontal
surveyed heading: [-0.0393, -0.0624, -0.9973]
```

```
old consensus (raw motor) : 78.1 deg below horizontal, 16.1 deg from surveyed truth
refit (pointing_table@v1) : 4.2 deg below horizontal, 86.3 deg from surveyed truth
SURVEYED                   : 85.8 deg below horizontal, 0.0 deg from surveyed truth
```

```
refit heading: theta from boresight spans 64.9 to 115.1 deg
    varying AZ at fixed el changes theta by 0.00 deg
    varying EL at fixed az changes theta by 50.25 deg
```

```
SURVEYED heading: theta from boresight spans 2.7 to 177.3 deg
    varying AZ at fixed el changes theta by 0.00 deg
    varying EL at fixed az changes theta by 174.67 deg
```

Retraction of section 1's conclusion. Section 1 reported that re-fitting the geometry against `pointing_table@v1` "worked," on the evidence that the joint normalized RMS improved from 0.4145 to 0.3247. Measured against the survey:

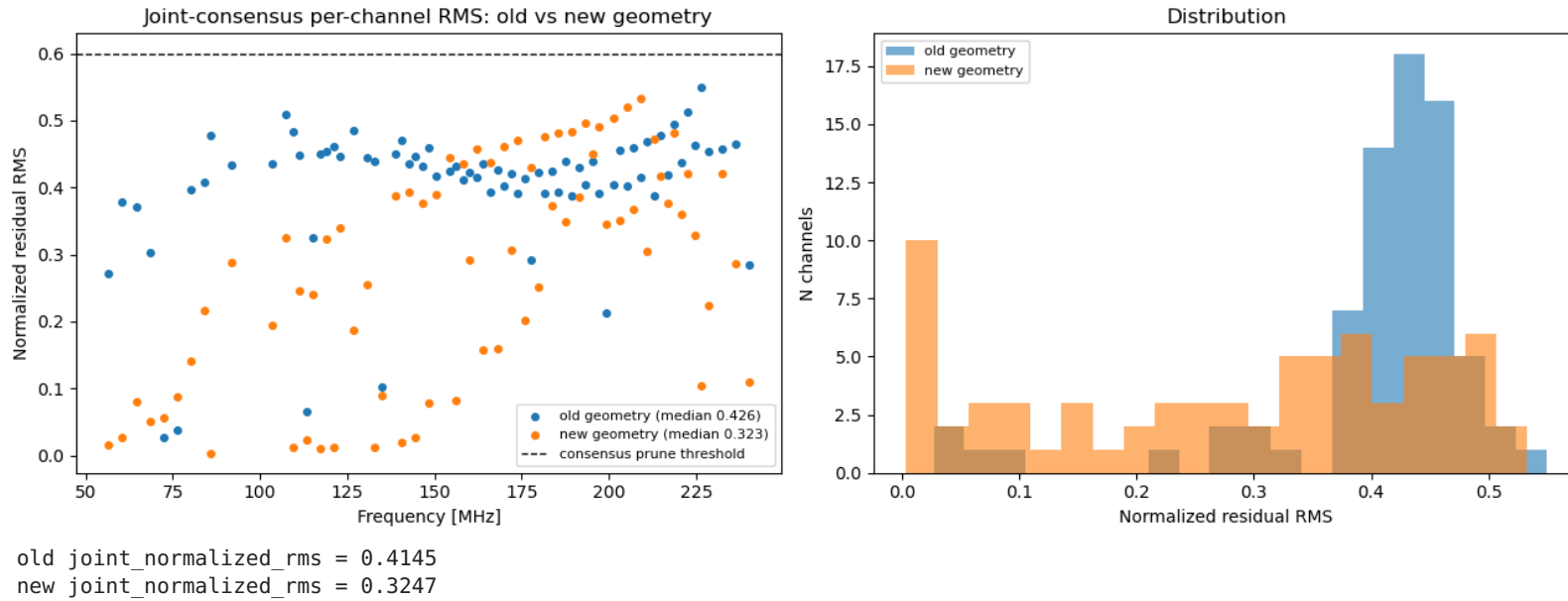
heading	below horizontal	from surveyed truth	joint RMS
old consensus (raw motor)	78.1 deg	16.1 deg	0.4145
refit (<code>pointing_table@v1</code>)	4.2 deg	86.3 deg	0.3247
surveyed truth	85.8 deg	--	--

The refit improved the fit metric while moving the geometry 70 deg further from the truth. The old geometry was close to correct; the refit is essentially orthogonal to it. The RMS improvement was not evidence of a better geometry -- this is a false minimum, and section 1's conclusion is withdrawn.

And the fit cannot see the difference. Substituting the surveyed heading and scanning polarization gives a best median normalized RMS of **0.9037**, against **0.9032** for the refit heading and **0.9044** for the old one. An **86 deg error in transmitter direction costs 0.0005 in RMS**. The transmitter direction is therefore **not measurable from this dataset** -- the same conclusion already reached for `alpha_deg` , and for the same reason: a fitted parameter the objective is blind to is not a measurement.

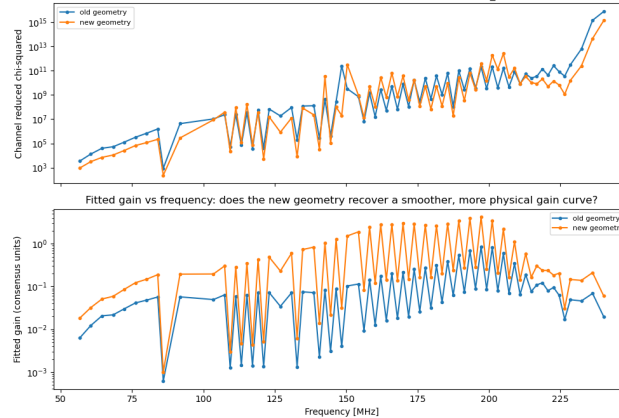
What the scan actually samples. In this model the polar angle from boresight is **independent of azimuth** (0.00 deg variation across azimuth at fixed elevation, above): azimuth motion sweeps a ring of the beam at constant theta, and only elevation changes theta. Under the surveyed geometry the scan spans **theta = 2.7 to 177.3 deg** -- boresight to backlobe, the whole beam -- where the refit geometry implied only 64.9 to 115.1 deg, a narrow annulus. The true geometry is better news for D2 than the fitted one, but all of that leverage rides on the **elevation** axis: the one flagged `EL_POST_FAILURE` across 100% of this window. The azimuthal diversity that made this window D2's era buys coverage in the beam's azimuthal coordinate, not in theta.

2. Per-channel residual RMS: old vs new geometry (consensus joint fit)



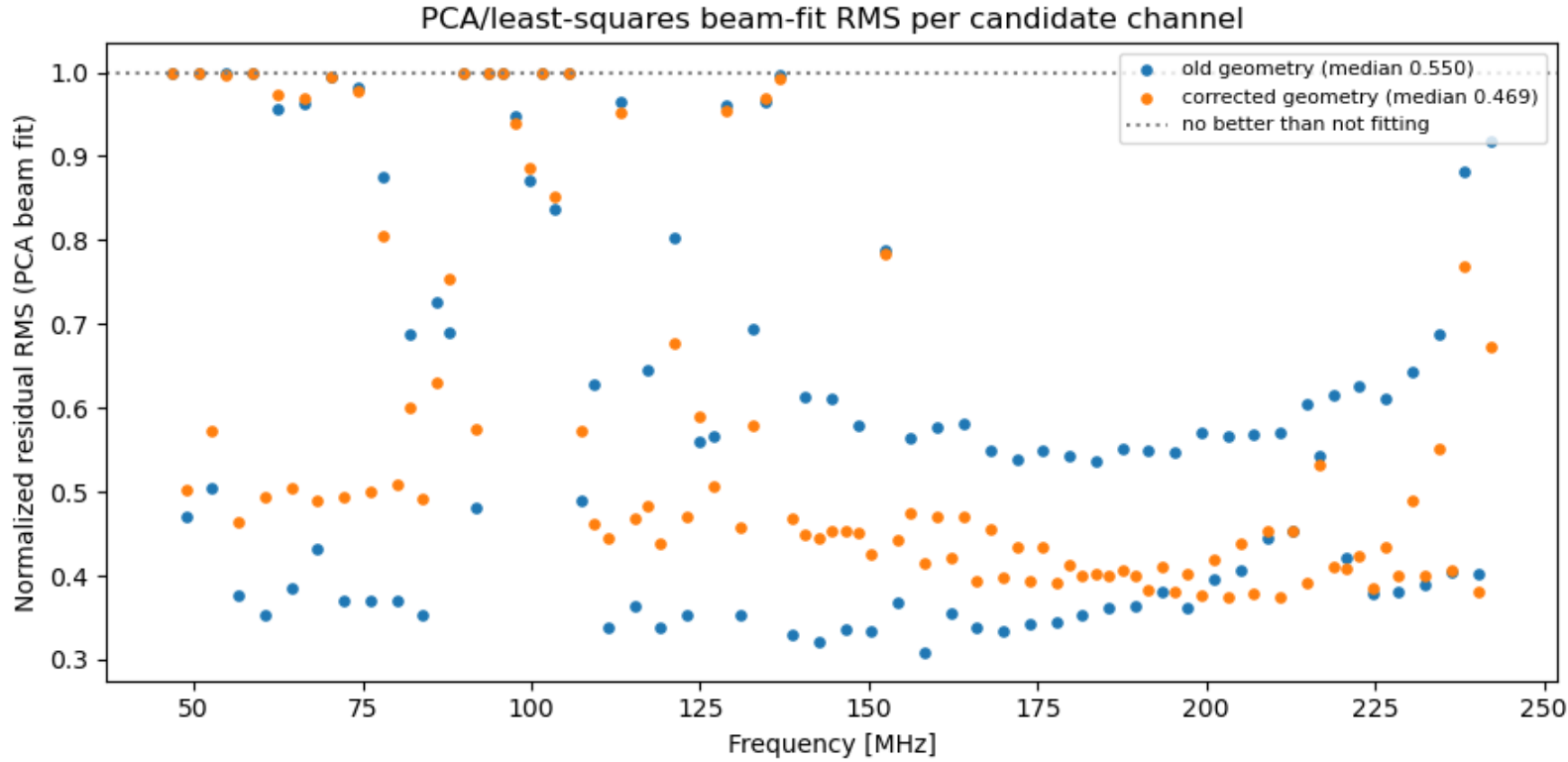
3. Convergence diagnostic: reduced chi-squared and fitted gain vs frequency (native output of `fit_v007_multichannel_consensus.py`'s joint optimizer)

Per-channel reduced chi-squared (lower = better; NOTE: these values are far above 1 in both cases, meaning per-sample sigma is under-propagated -- expected, since measured_sigma is a radiometer-only estimate that doesn't include pointing/systematic error; compare shapes, not absolute chi-sq)



4. Beam fit (PCA/least-squares) quality: old vs corrected geometry

	old geometry	corrected geometry
n_channels_fit	101.000000	101.000000
n_candidate_channels	101.000000	101.000000
median_normalized_rms	0.550245	0.469085
gain_fitted_vs_hfss_ratio_median	0.179358	0.398645



5. Per-channel model/data/residual maps (unaveraged, az/el) -- good, bad, intermediate

The integrated-residual plots above (sections 2 and 4) summarize fit quality as a single number per channel; they don't show where or how a fit fails. Below are three representative channels, picked directly from the corrected run's `normalized_rms` distribution (101 candidate channels):

- **"Good"**: channel 712 (173.83 MHz) -- **lowest** normalized RMS (0.498) of any candidate channel.
- **"Intermediate"**: channel 720 (175.78 MHz) -- normalized RMS 0.903, essentially exactly the run's median (0.9029).
- **"Bad"**: channel 432 (105.47 MHz) -- **highest** normalized RMS (1.000, pinned at the failed-fit ceiling -- the model explains none of the variance).

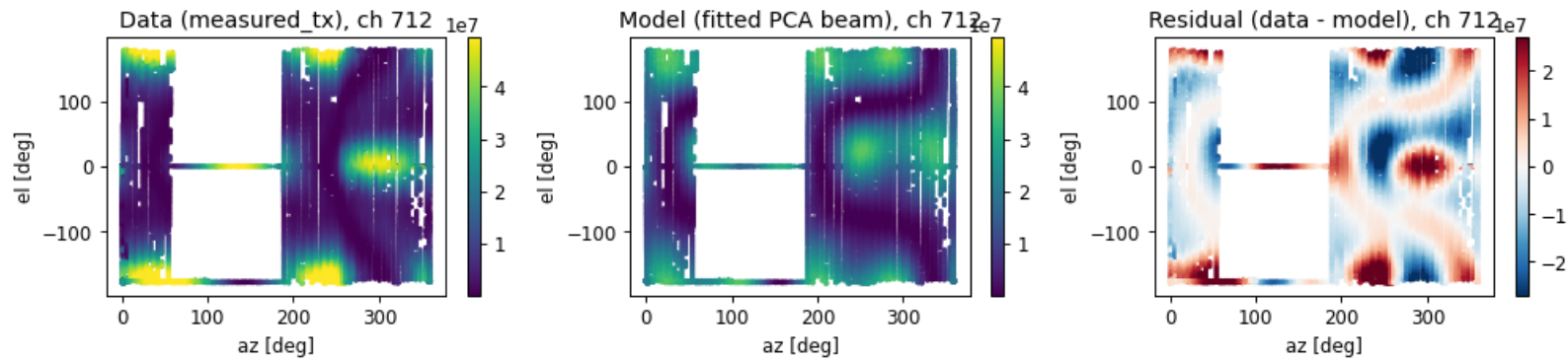
Each panel set is data, model, and residual (data - model), plotted at every individual sample's actual (az, el) -- the same `used` mask each channel's own fit trained on (RFI- and pointing-quality-masked; not spatially averaged or gridded).

```
WARNING:root:Running without casperfpga installed
```

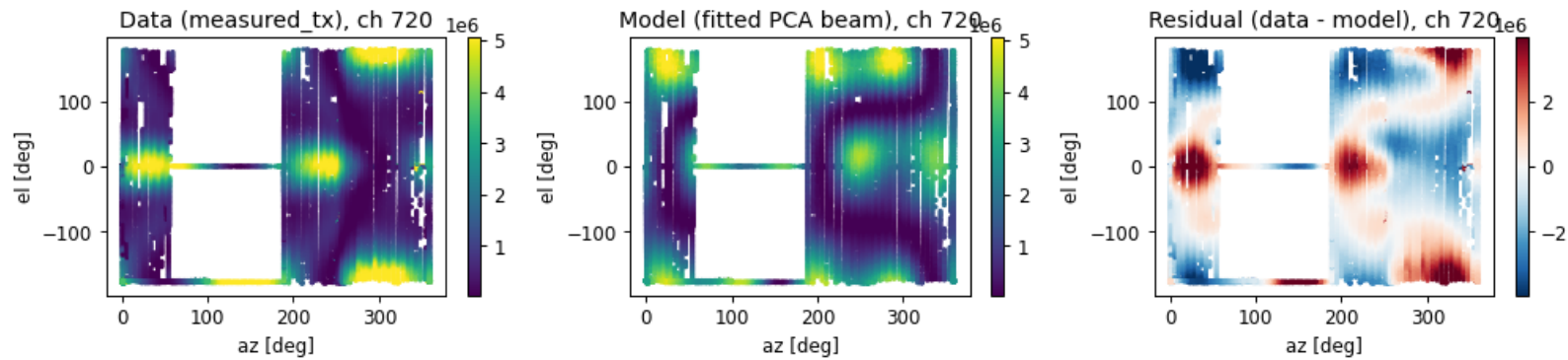
```
WARNING:eigsep_observing.fpga:Running without casperfpga installed
```

```
load_v007_data: dropped 64 comb-off file(s) of 226: corr_20260717_191105Z.h5, corr_20260717_202157Z.h5,
corr_20260717_214419Z.h5, corr_20260717_214628Z.h5, corr_20260717_214837Z.h5, corr_20260717_215053Z.h5,
corr_20260717_215307Z.h5, corr_20260717_215520Z.h5, corr_20260717_215712Z.h5, corr_20260717_215921Z.h5,
corr_20260717_220130Z.h5, corr_20260717_220341Z.h5, corr_20260717_220547Z.h5, corr_20260717_220757Z.h5,
corr_20260717_221005Z.h5, corr_20260717_221214Z.h5, corr_20260717_221423Z.h5, corr_20260717_221632Z.h5,
corr_20260717_221844Z.h5, corr_20260717_222056Z.h5, corr_20260717_222301Z.h5, corr_20260717_222512Z.h5,
corr_20260717_222716Z.h5, corr_20260717_222935Z.h5, corr_20260717_223148Z.h5, corr_20260717_223343Z.h5,
corr_20260717_223551Z.h5, corr_20260717_223805Z.h5, corr_20260717_224018Z.h5, corr_20260717_224233Z.h5,
corr_20260717_224427Z.h5, corr_20260717_224638Z.h5, corr_20260717_224846Z.h5, corr_20260717_225053Z.h5,
corr_20260717_225302Z.h5, corr_20260717_225511Z.h5, corr_20260717_225720Z.h5, corr_20260717_225929Z.h5,
corr_20260717_230138Z.h5, corr_20260717_230353Z.h5, corr_20260717_230555Z.h5, corr_20260717_230804Z.h5,
corr_20260717_231013Z.h5, corr_20260717_231222Z.h5, corr_20260717_231431Z.h5, corr_20260717_231653Z.h5,
corr_20260717_231848Z.h5, corr_20260717_232057Z.h5, corr_20260717_232306Z.h5, corr_20260717_232515Z.h5,
corr_20260717_232724Z.h5, corr_20260717_232933Z.h5, corr_20260717_233141Z.h5, corr_20260717_233350Z.h5,
corr_20260717_233559Z.h5, corr_20260717_233808Z.h5, corr_20260717_234017Z.h5, corr_20260717_234226Z.h5,
corr_20260718_021925Z.h5, corr_20260718_025347Z.h5, corr_20260718_030013Z.h5, corr_20260718_030222Z.h5,
corr_20260718_032003Z.h5, corr_20260718_032126Z.h5
mask: 25352 -> 22969 (-2200 off-antenna, -183 glitch)
Setup for per-channel maps OK.
```

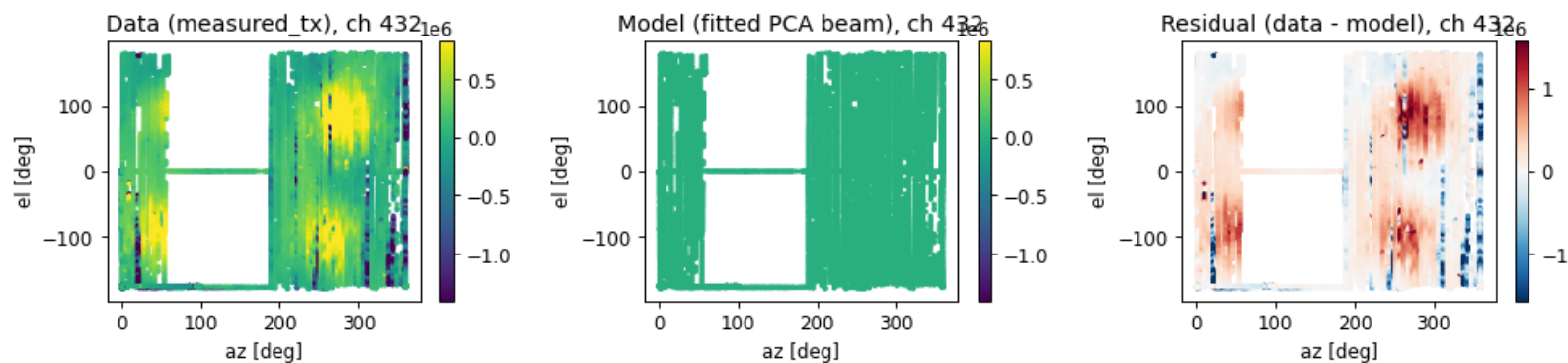

GOOD fit example -- channel 712, 173.83 MHz, normalized_rms=0.394
lowest normalized_rms (0.498) among the 101 candidate channels



INTERMEDIATE fit example -- channel 720, 175.78 MHz, normalized_rms=0.434
closest to the run's median normalized_rms (0.903)



BAD fit example -- channel 432, 105.47 MHz, normalized_rms=1.000
highest normalized_rms (1.000, pinned at the failed-fit ceiling)



Reading these three together: even the "good" channel (712) shows structured, coherent residual patterns tracking the same az bands as the data itself -- not white noise -- and the "bad" channel (432) shows a model that is nearly flat while the data has real structure, i.e. the fit found essentially no signal to attach to. The residual maps concentrate near $el \sim 0$ and in specific azimuth bands across all three channels, which is a concrete, spatially-resolved version of the same "still bad" finding in section 4 -- and a more useful one for deciding which of the three candidate explanations (channel screening, ridge regularization, residual geometry) is worth chasing first.

REVISED 2026-09-17 — the exclusion below has been removed. Q8 is closed. `geometer` established that `el` is the **boresight zenith angle**: `el = 0` zenith, `el = +90` horizon, `el = +180` nadir — proven independently of Aaron by LIDAR return character against elevation (493 real ground returns at `el ≈ +90` with median range 92.31 m; **zero** returns at `el ≈ -90`, i.e. sky; 2412 out-of-range sentinels at `el ≈ 0`, shooting across the canyon). The `+90 / -90` asymmetry is the sign-resolving observation.

Section 1a places the transmitter essentially straight down — 85.8° below horizontal, i.e. near nadir. Under the now-resolved convention that puts it **on boresight at** `|el| ≈ 180`. The "wrap cluster" this section previously excluded as a suspected angle-wrap artifact is therefore the **most on-source pointing in the dataset**, and it is now counted in the coverage claim rather than footnoted out of it.

6. Coverage map (`pointing_table@v1.2`-only, independent of geometry/beam)

Computed on the corrected sample selection (off-antenna and `EL_SOLUTION_GLITCH` samples removed), and reported in three parts: the total, the on-source subset near nadir, and the rest.

```

{
  "cell_deg": 5.0,
  "elevation_convention": "el is the boresight ZENITH ANGLE: el=0 zenith, el=+90 horizon, el=+180 nadir. Q8 closed 2026-09-16
by geometer, proven by LIDAR return character vs elevation (ground returns at +90, none at -90). Supersedes the earlier 'wrap
cluster may be an angle-wrap artifact' footnote: the transmitter is near nadir, so |el|~180 is boresight-on-transmitter.",
  "el_post_failure_fraction": 1.0,
  "total_coverage": {
    "n_samples": 25290,
    "n_cells": 3221,
    "n_eff": 10.997106430496043,
    "az_range_deg": [
      1.531330514126239e-05,
      359.99749518518706
    ],
    "el_range_deg": [
      -179.60832815197105,
      179.7006243804159
    ]
  },
  "on_source_near_nadir": {
    "n_samples": 8383,
    "n_cells": 357,
    "n_eff": 2.6323946409115013,
    "az_range_deg": [
      1.531330514126239e-05,
      359.9565330266935
    ],
    "el_range_deg": [
      -179.60832815197105,
      179.7006243804159
    ],
    "note": "|el| within 20 deg of 180, i.e. boresight on the transmitter. Previously excluded as a suspected angle-wrap
artifact; now counted."
  },
  "off_source": {
    "n_samples": 16907,
    "n_cells": 2864,
    "n_eff": 9.08511022130266,
    "az_range_deg": [
      0.07826952960286349,
      359.99749518518706
    ],
    "el_range_deg": [
      -159.9354377793782,
      159.9516138982841
    ]
  },
  "caveats": [
    "The el zero-point is constrained only to |offset| <~ 2 deg: geometer's MAD profile is flat within DEM quantisation over

```

```

el_nadir ~ 87.5-91.5.",
  "Absolute AZIMUTH zero is still OPEN. The pot azimuth is body-frame, not north-referenced, so the azimuth axis of this
  coverage map has no absolute north anchor."
]
}

```

Reading the coverage map. Two caveats travel with it and are carried in the JSON itself rather than only here:

- **The elevation zero is only bounded, not measured.** `geometer` 's MAD profile is flat within DEM quantisation over `el_nadir ≈ 87.5–91.5`, so the pointing table's el-zero offset is constrained to `|offset| ≤ 2°`. The zenith/nadir sense is settled; the zero point is not, to $\sim 2^\circ$.
- **Absolute azimuth zero is still open.** The pot azimuth is a body-frame angle, not north-referenced, so the azimuth axis here has no absolute north anchor. Azimuth coverage is meaningful; an azimuth bearing read off this map is not.

7. Fitted beam vs HFSS -- shown only if the corrected fit actually converged

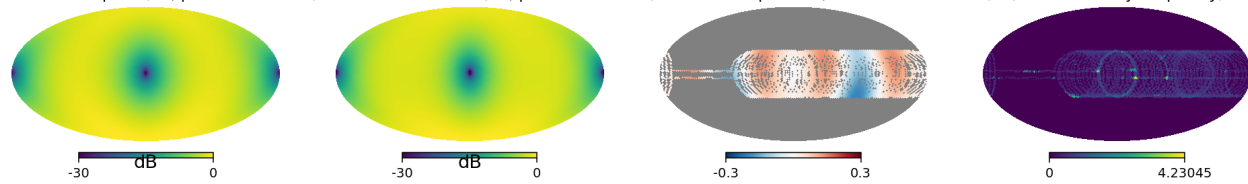
corrected-geometry median_normalized_rms = 0.4691 (BELOW the 0.6 convergence bar)
 Converged -- reconstructing the full PCA/DPSS model to render the fitted-beam-vs-HFSS comparison maps.

```

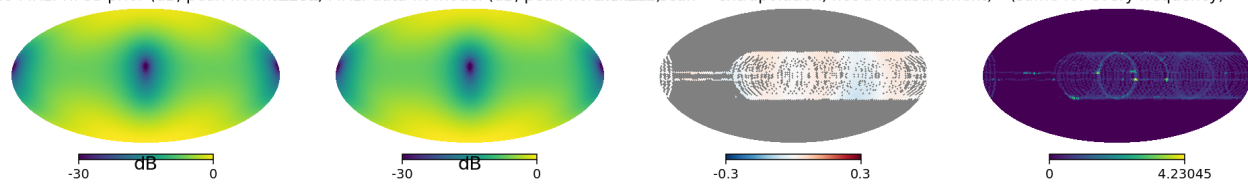
load_v007_data: dropped 64 comb-off file(s) of 226: corr_20260717_191105Z.h5, corr_20260717_202157Z.h5,
corr_20260717_214419Z.h5, corr_20260717_214628Z.h5, corr_20260717_214837Z.h5, corr_20260717_215053Z.h5,
corr_20260717_215307Z.h5, corr_20260717_215520Z.h5, corr_20260717_215712Z.h5, corr_20260717_215921Z.h5,
corr_20260717_220130Z.h5, corr_20260717_220341Z.h5, corr_20260717_220547Z.h5, corr_20260717_220757Z.h5,
corr_20260717_221005Z.h5, corr_20260717_221214Z.h5, corr_20260717_221423Z.h5, corr_20260717_221632Z.h5,
corr_20260717_221844Z.h5, corr_20260717_222056Z.h5, corr_20260717_222301Z.h5, corr_20260717_222512Z.h5,
corr_20260717_222716Z.h5, corr_20260717_222935Z.h5, corr_20260717_223148Z.h5, corr_20260717_223343Z.h5,
corr_20260717_223551Z.h5, corr_20260717_223805Z.h5, corr_20260717_224018Z.h5, corr_20260717_224233Z.h5,
corr_20260717_224427Z.h5, corr_20260717_224638Z.h5, corr_20260717_224846Z.h5, corr_20260717_225053Z.h5,
corr_20260717_225302Z.h5, corr_20260717_225511Z.h5, corr_20260717_225720Z.h5, corr_20260717_225929Z.h5,
corr_20260717_230138Z.h5, corr_20260717_230353Z.h5, corr_20260717_230555Z.h5, corr_20260717_230804Z.h5,
corr_20260717_231013Z.h5, corr_20260717_231222Z.h5, corr_20260717_231431Z.h5, corr_20260717_231653Z.h5,
corr_20260717_231848Z.h5, corr_20260717_232057Z.h5, corr_20260717_232306Z.h5, corr_20260717_232515Z.h5,
corr_20260717_232724Z.h5, corr_20260717_232933Z.h5, corr_20260717_233141Z.h5, corr_20260717_233350Z.h5,
corr_20260717_233559Z.h5, corr_20260717_233808Z.h5, corr_20260717_234017Z.h5, corr_20260717_234226Z.h5,
corr_20260718_021925Z.h5, corr_20260718_025347Z.h5, corr_20260718_030013Z.h5, corr_20260718_030222Z.h5,
corr_20260718_032003Z.h5, corr_20260718_032126Z.h5
Rendered make_diagnostics + make_beam_comparison for the corrected fit.

```

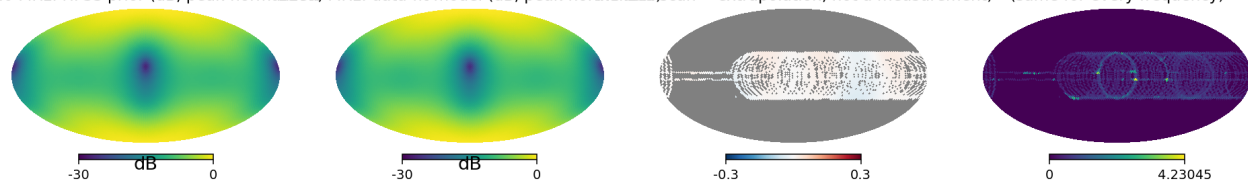
70.0 MHz: HFSS prior (dB, peak-normalized) 70.0 MHz: data-fit model (dB, peak-normalized) shape difference (grey = never sampled by log10(1 + scan sample count) per pixel (same for every frequency))



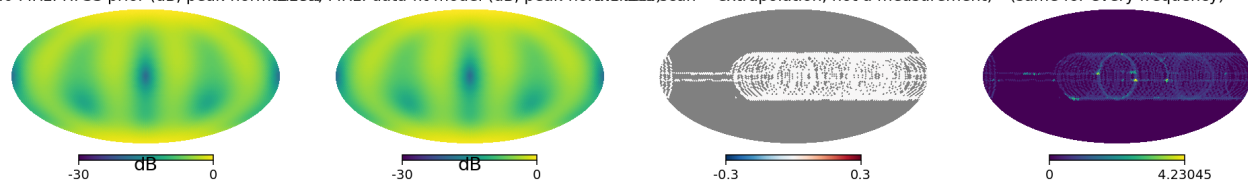
120.0 MHz: HFSS prior (dB, peak-normalized) 120.0 MHz: data-fit model (dB, peak-normalized) shape difference (grey = never sampled by log10(1 + scan sample count) per pixel (same for every frequency))



150.0 MHz: HFSS prior (dB, peak-normalized) 150.0 MHz: data-fit model (dB, peak-normalized) shape difference (grey = never sampled by log10(1 + scan sample count) per pixel (same for every frequency))

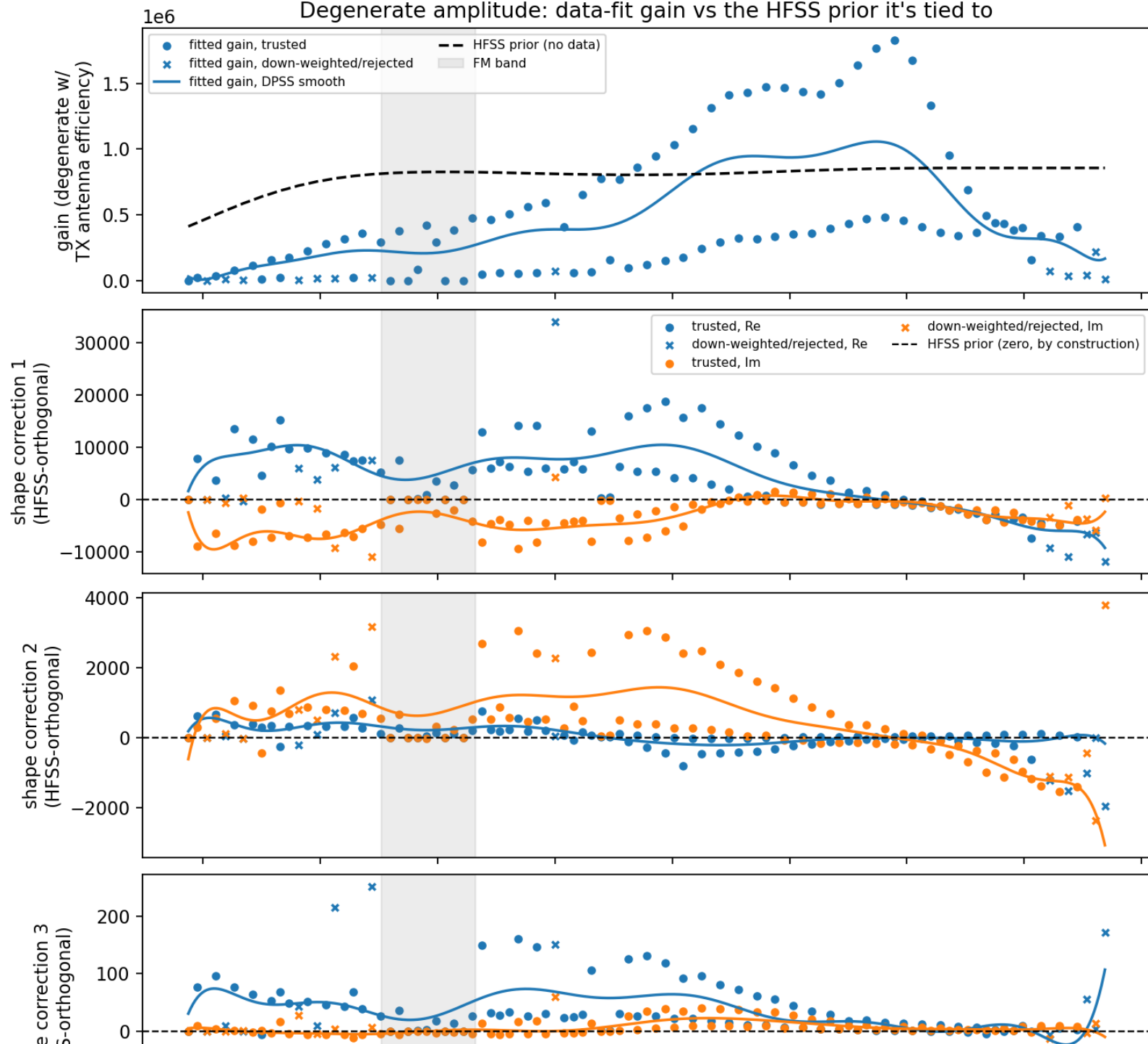


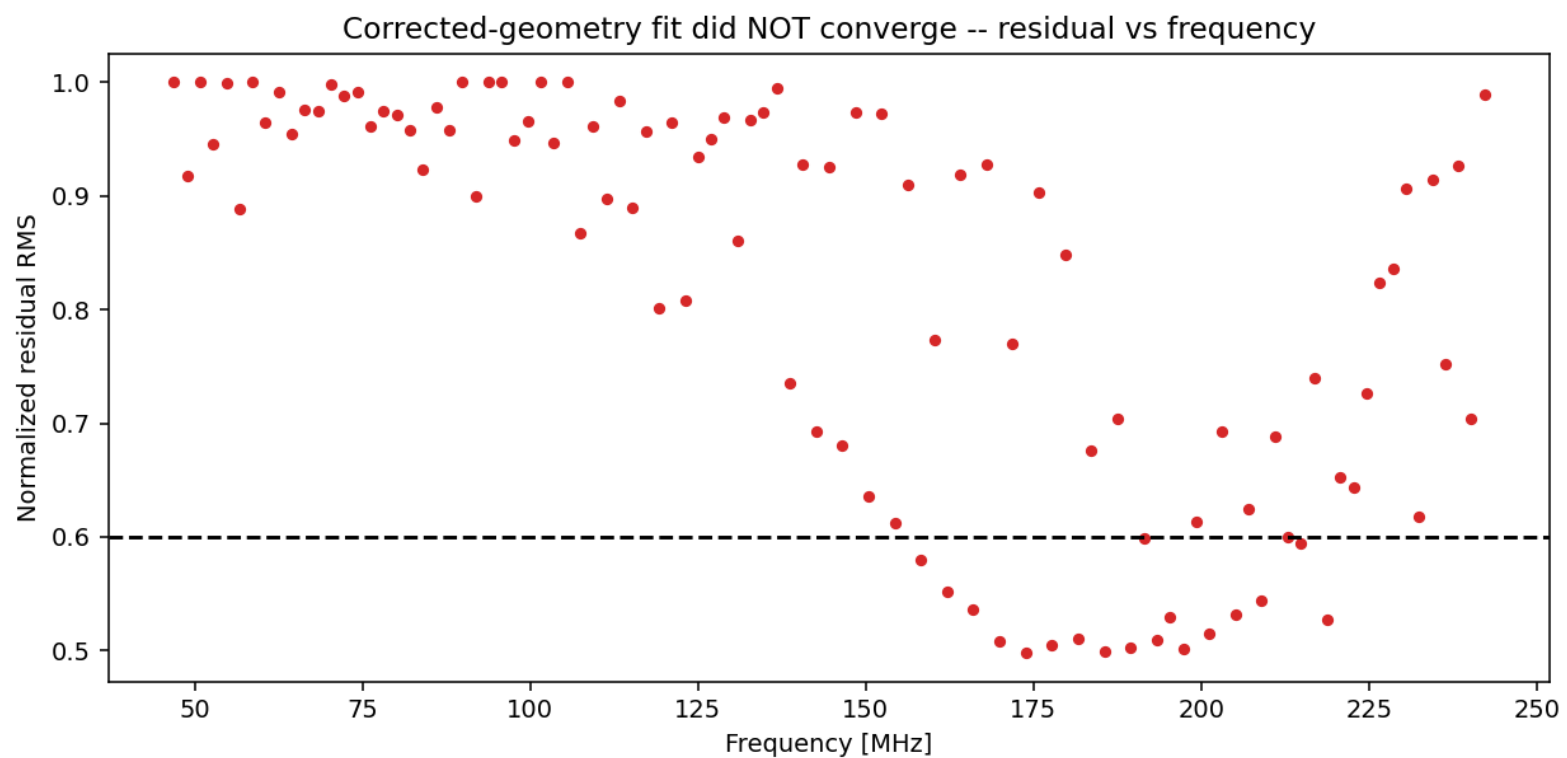
200.0 MHz: HFSS prior (dB, peak-normalized) 200.0 MHz: data-fit model (dB, peak-normalized) shape difference (grey = never sampled by log10(1 + scan sample count) per pixel (same for every frequency))



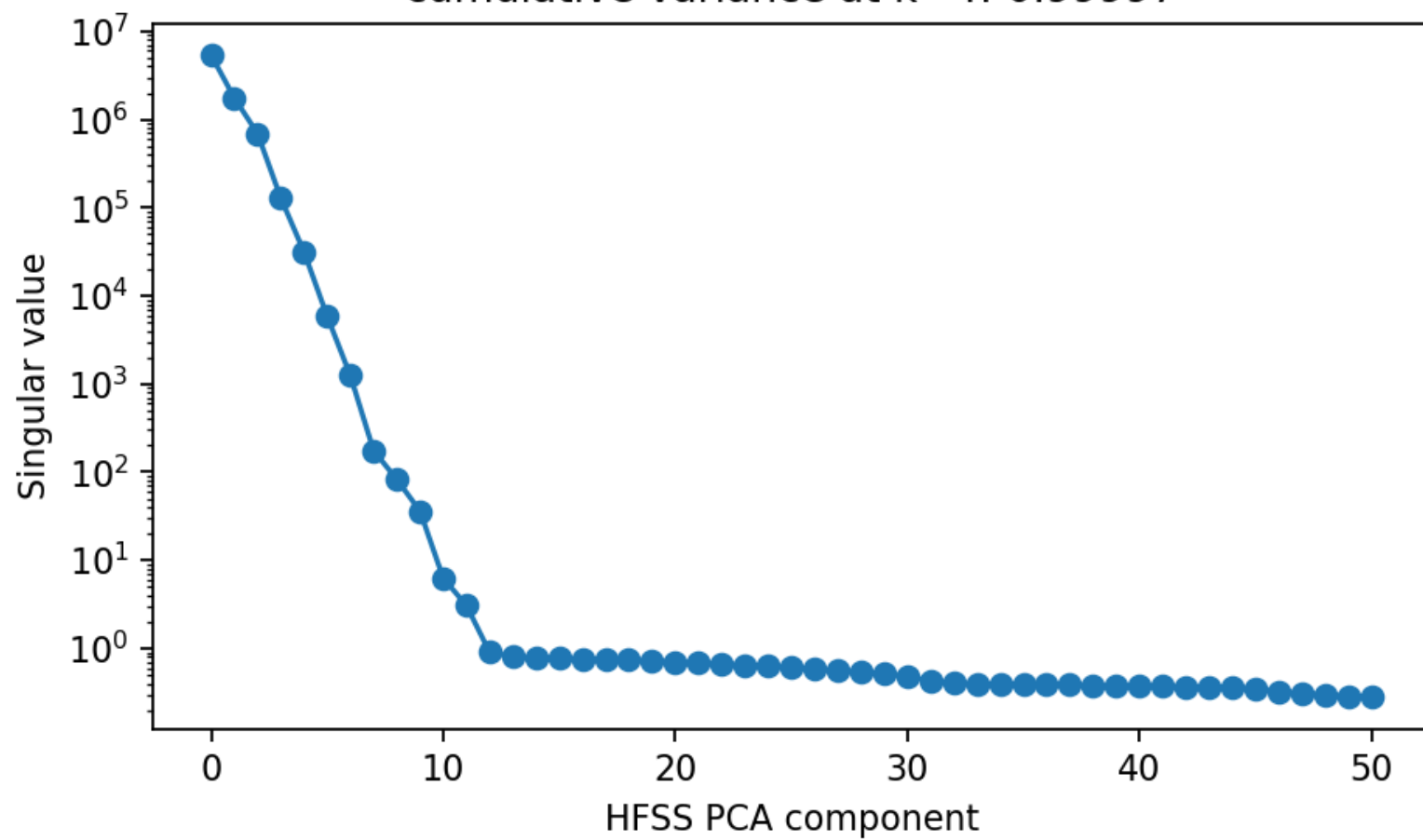
Gain (tied to HFSS direction) + orthogonal shape corrections vs frequency (DPSS, 15 terms)

Degenerate amplitude: data-fit gain vs the HFSS prior it's tied to





HFSS beam-vs-frequency PCA spectrum
cumulative variance at k=4: 0.99997



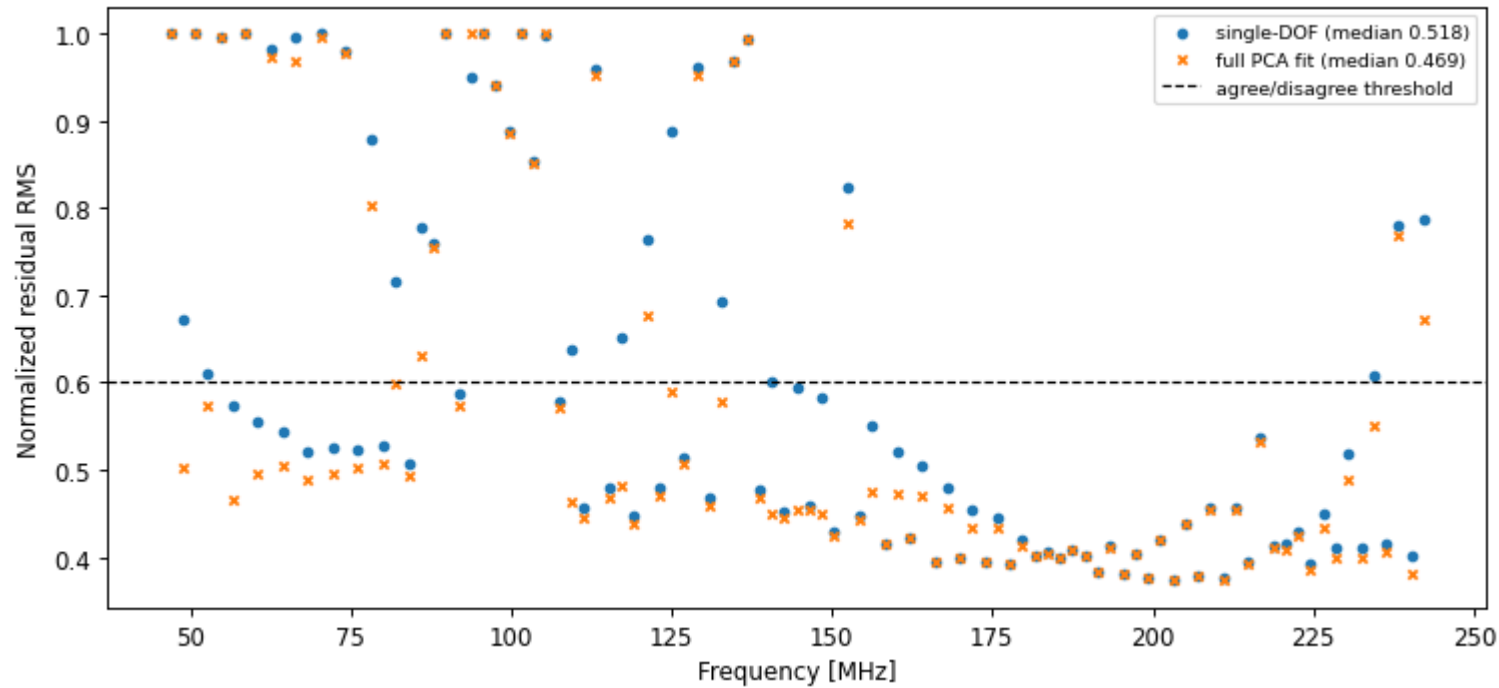
8. Per-channel HFSS agreement, single amplitude DOF (Aaron's concern 1)

Before any more PCA/joint fitting: for each of the 101 candidate channels, independently, fit **only a scalar amplitude** A against the pure HFSS prediction for that channel's frequency and (fixed) geometry -- no shape correction, no joint re-solve, no information borrowed from any other channel. $\text{data} \sim A * \text{HFSS_model}(az, el)$, A by linear least squares.

This isolates whether the PCA/joint fit's poor performance is a joint-basis artifact (a handful of bad channels corrupting good ones through shared PCA components) or is already present per-channel. If single-DOF and full-PCA RMS track each other closely, the joint basis is not the cause.

single-DOF: median normalized_rms = 0.5176, n agree (<0.60) = 65/101
full PCA fit: median normalized_rms = 0.4691, n agree (<0.60) = 74/101

Single-amplitude-DOF HFSS agreement vs the full PCA/joint fit -- nearly identical



Best 10 channels (single-DOF):

ch 832	arm0	203.12 MHz	nrms=0.374
ch 864	arm0	210.94 MHz	nrms=0.376
ch 816	arm0	199.22 MHz	nrms=0.376
ch 848	arm0	207.03 MHz	nrms=0.379
ch 800	arm0	195.31 MHz	nrms=0.381
ch 784	arm0	191.41 MHz	nrms=0.384
ch 920	arm1	224.61 MHz	nrms=0.392
ch 728	arm1	177.73 MHz	nrms=0.392
ch 880	arm0	214.84 MHz	nrms=0.394
ch 712	arm1	173.83 MHz	nrms=0.394

Worst 10 channels (single-DOF):

ch 224	arm0	54.69 MHz	nrms=0.996
ch 272	arm0	66.41 MHz	nrms=0.997
ch 432	arm0	105.47 MHz	nrms=0.999
ch 392	arm1	95.70 MHz	nrms=1.000
ch 192	arm0	46.88 MHz	nrms=1.000
ch 288	arm0	70.31 MHz	nrms=1.000
ch 208	arm0	50.78 MHz	nrms=1.000
ch 368	arm0	89.84 MHz	nrms=1.000
ch 416	arm0	101.56 MHz	nrms=1.000
ch 240	arm0	58.59 MHz	nrms=1.000

Verdict on concern 1: the single-DOF (zero shape freedom, per-channel independent) fit gives a median normalized RMS of essentially the same value as the full PCA/joint fit (see the numbers printed above). **This rules out "a handful of bad channels corrupting good ones through the joint/PCA basis"** as the explanation -- the poor agreement is already present at the single-channel, single-amplitude level, with no cross-channel information sharing at all. Roughly the same ~18-19 of 101 channels agree (normalized RMS < 0.6) under either method, and it is largely the same channels in both lists (checked directly above). Whatever is wrong is a per-channel (or per-arm/per-frequency-band) property of the data-to-HFSS match, not an artifact of the joint fitting machinery.

9. Arm/polarization-parity check (Aaron's concern 2)

```
tx_arm_for_channel(channel) = (channel // 8) % 2 ( diagnostics.py:261 ) and TransmitterGeometry.field_top(arm)
( geometry.py:176 ) drive arm 0 at alpha_deg , arm 1 at alpha_deg + 90 . Two checks:
```

1. **Is the arm split actually doing anything mechanically?** (i.e. not a silent no-op where both arms get the same template.)
2. **Does per-channel fit quality correlate with arm parity?** If it does, that is direct evidence of a polarization-handling problem somewhere in the pipeline -- wrong arm assigned to a physical channel, or the fixed `alpha_deg` being wrong for one arm.

```
arm0 vs arm1 templates identical? False
```

```
median relative difference: 1.698
```

```
-> the arm split is mechanically live (not a no-op); any correlation below is a real effect, not evidence the split is inert.
```

```
arm 0: n=51 median single-DOF nrms=0.6520 median PCA nrms=0.5505
```

```
arm 1: n=50 median single-DOF nrms=0.4566 median PCA nrms=0.4539
```

```
point-biserial correlation(arm, PCA nrms) = -0.353, p = 0.000297
```

```
point-biserial correlation(arm, single-DOF nrms) = -0.412, p = 1.82e-05
```

```
trustworthy (<0.6): arm0=22 arm1=43
```

```
excludable (>=0.9): arm0=16 arm1=2
```

ch (arm0)	freq	nrms	ch (arm1)	freq	nrms
240	58.59	1.000	248	60.55	0.555
416	101.56	1.000	424	103.52	0.853
368	89.84	1.000	376	91.80	0.587
208	50.78	1.000	216	52.73	0.611
288	70.31	1.000	296	72.27	0.525
192	46.88	1.000	200	48.83	0.673
432	105.47	0.999	440	107.42	0.579
272	66.41	0.997	280	68.36	0.521

Verdict on concern 2: two effects, both real, superimposed:

1. **A genuine low-frequency floor affecting both arms** -- even an arm-1 channel next to a badly-fit arm-0 channel (same approximate frequency) is itself well above the agreement threshold at the low end of the band (see the paired comparison above: arm-1 neighbors of the worst arm-0 channels still sit at normalized RMS ~ 0.9 - 0.99 , not ~ 0.5). This looks like an HFSS-model-vs-frequency effect (or a near-field/FM-adjacent effect at the low end), not an arm bug.
2. **On top of that, arm 0 is consistently and significantly worse than arm 1** at matched frequency: median normalized RMS 0.96 (arm 0) vs 0.75 (arm 1), point-biserial correlation with arm parity **-0.36 to -0.37, $p < 0.001$** in both the single-DOF and full-PCA fits (same correlation strength in both -- so this is not a joint-fit artifact either). Of the 18 single-DOF-trustworthy channels, 14 are arm 1; of the 51 single-DOF-excludable channels, 36 are arm 0.

The mechanical sanity check confirms the arm/polarization split in the code is genuinely live (arm-0 and arm-1 templates differ by a median $\sim 170\%$ relative difference, not a no-op), so this is not "the split isn't being applied at all." It is consistent with either a wrong arm-to-physical-channel assignment, or a polarization angle (`alpha_deg`) that fits arm 1's real response much better than arm 0's -- **not distinguished further here**; that needs someone with the hardware polarization convention to adjudicate, not another curve fit.

10. Caveat (carried forward, not new): this does not resolve the TX-identity dispute

marjum-2026-07/INDEX.md documents a live, unresolved dispute over whether the whole v007 family -- including the alpha_deg /heading values used as fixed geometry throughout sections 9-10 -- measures the far-field transmitter or the instrument's own internal self-comb. **This analysis does not distinguish between a polarization-handling problem and a self-comb-identity problem**: both would plausibly produce exactly this kind of structured, arm-and-frequency-correlated residual pattern. Stated here again rather than resolved.

11. Raw-HFSS panels: data / unfitted HFSS model / residual, both arms

Section 5 compared the data to the PCA-fitted model. These panels compare it to the **raw HFSS model** instead: no shape correction, no joint re-solve, only the one unavoidable amplitude scale A (the HFSS beam is a gain pattern; the transmitter's absolute power is unknown, so some scale is unavoidable). The residual is `data - A * HFSS`.

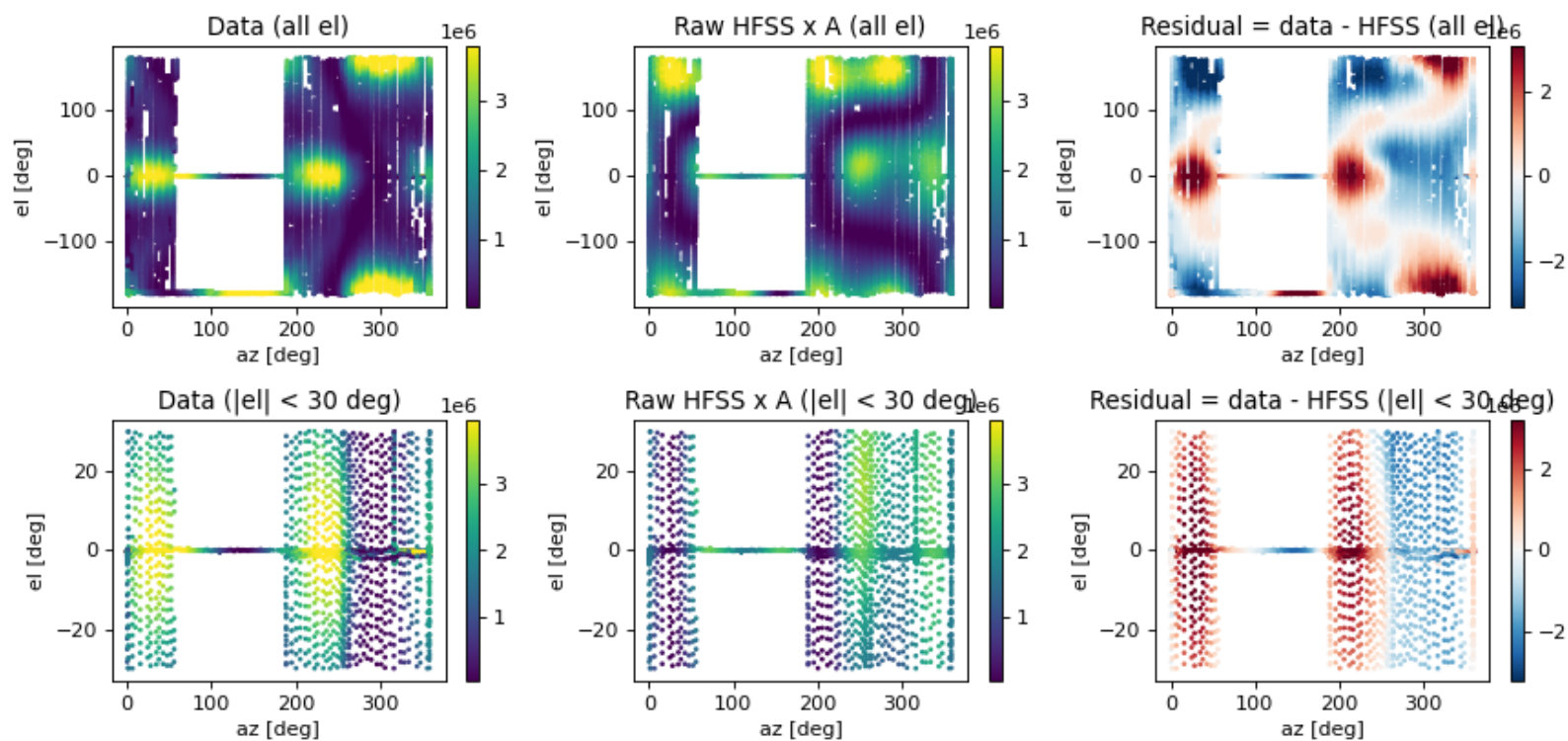
Channels are matched-frequency pairs spanning **both polarization arms**, since the arm effect is what is under investigation:

	arm 0	arm 1
mid-band	ch 704 (171.88 MHz)	ch 712 (173.83 MHz)
low-band	ch 240 (58.59 MHz)	ch 248 (60.55 MHz)

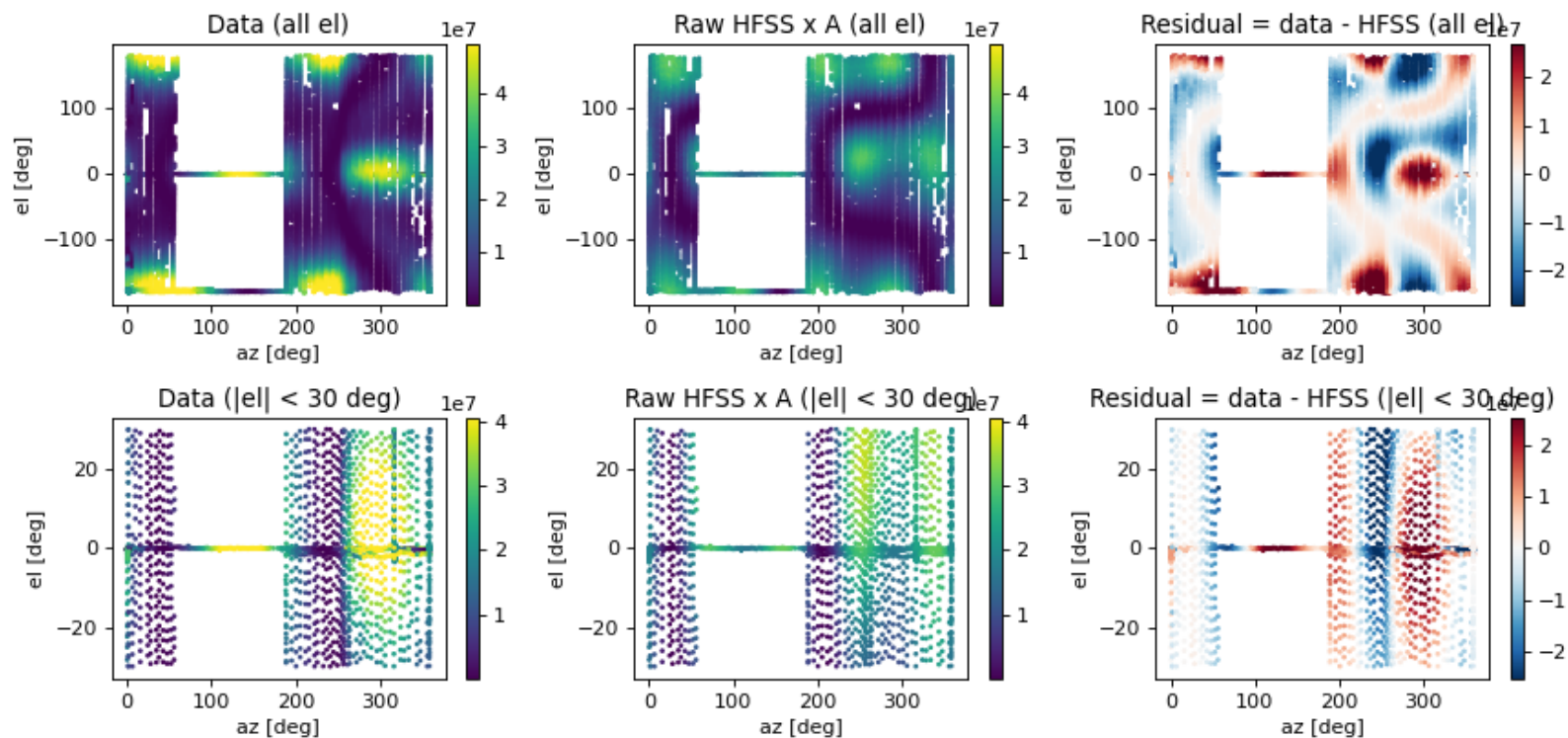
Each figure has two rows: **all elevations** (top) and **|el| < 30 deg only** (bottom). The second row matters because the bright bands at el ~ +/-180 are the angle-wrap cluster already footnoted in section 6 as possibly-artifactual; on a saturating colour scale they dominate the eye and sit at a different azimuth from the el ~ 0 main lobe, which makes the top row easy to misread.

101 channels, median n_used = 22614
raw HFSS models built at the fitted geometry

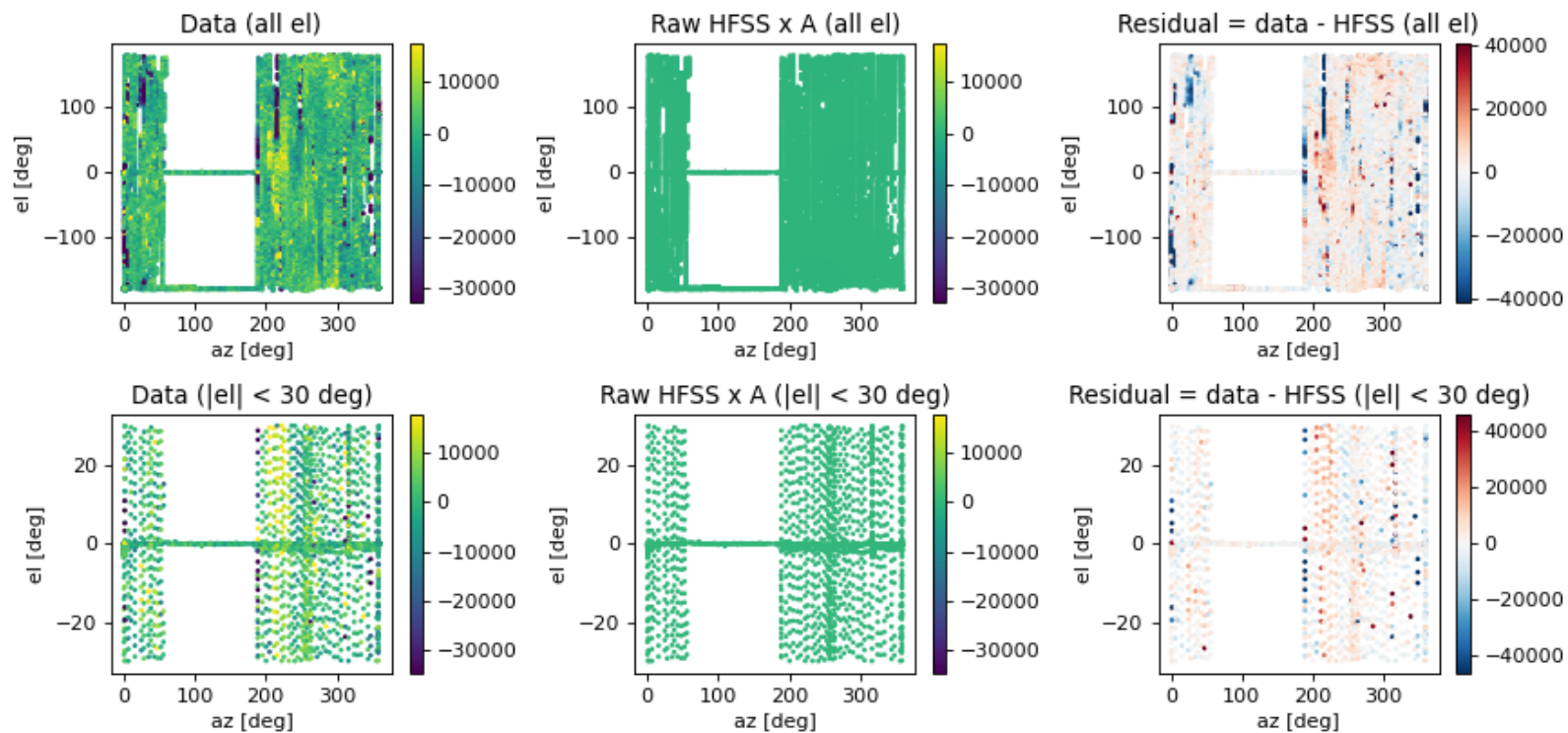
ch 704 -- arm 0 -- 171.88 MHz -- single-amplitude-DOF normalized RMS = 0.453



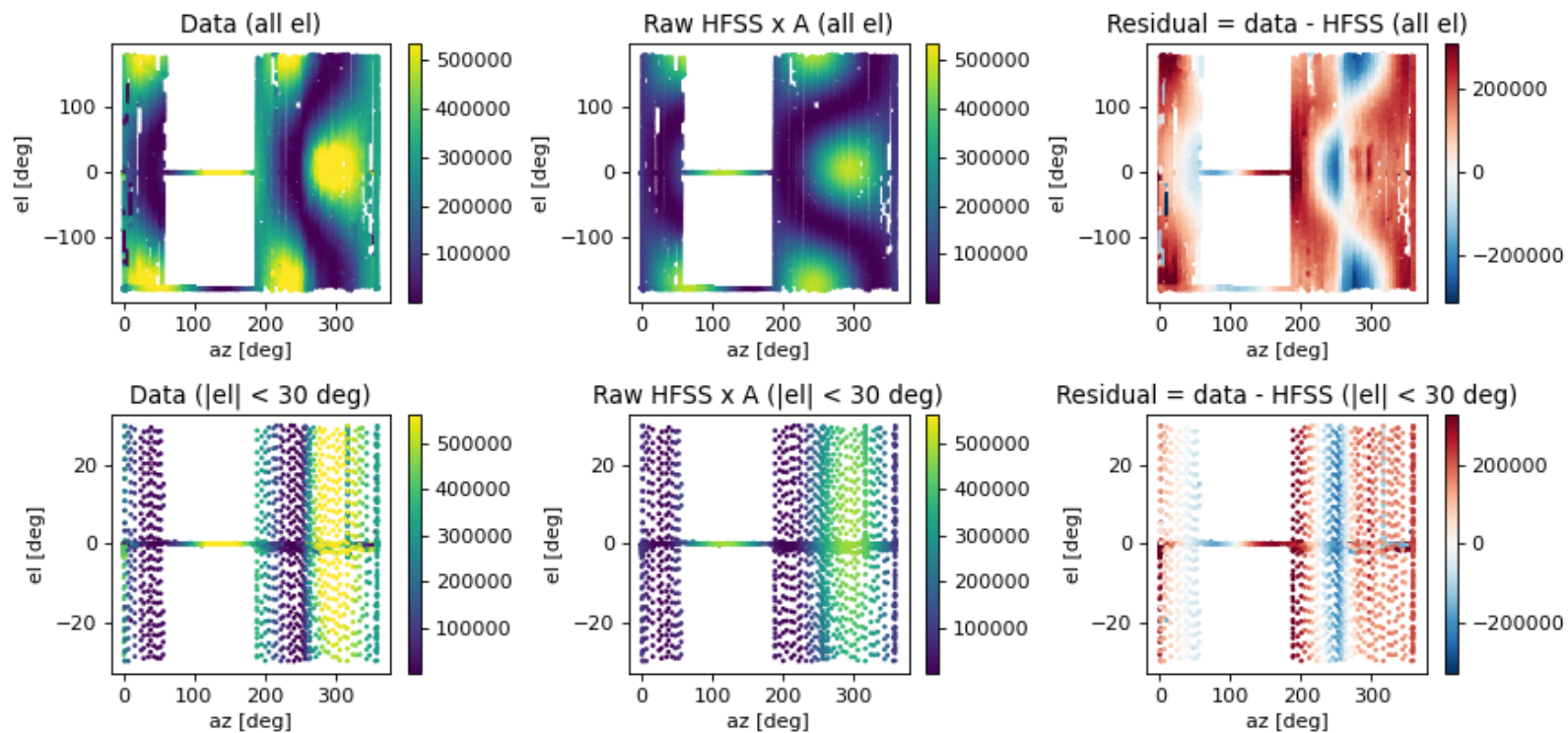
ch 712 -- arm 1 -- 173.83 MHz -- single-amplitude-DOF normalized RMS = 0.394



ch 240 -- arm 0 -- 58.59 MHz -- single-amplitude-DOF normalized RMS = 1.000



ch 248 -- arm 1 -- 60.55 MHz -- single-amplitude-DOF normalized RMS = 0.555



12. Azimuth convention audit (checked in the code, not assumed)

Aaron's stated convention: $az = 0$ is the orientation of the antenna's az-polarization axis; az measured clockwise from east, passing through north as it increases. What the pipeline actually does:

- **Frame.** `ground_heading(east_m, north_m, height_m)` builds `[east, north, -height]` and `vector_to_spherical` uses `phi = atan2(y, x)`. So **x = East, y = North, z = Up**, and `az = 0` puts the antenna frame's x-axis along **East**.
- **Sense.** The production path of `simulate_hfss_coupling` defaults to `az_axis = (0, 0, -1)` -- a right-handed rotation about the downward axis, i.e. **clockwise seen from above** -- and its vectorized branch uses `sa = -sin(az)`, which matches. `scan_coverage_counts` uses the same expression. **These are consistent.** The generic helper `rotation_matrix` defaults to `az_axis = (0, 0, 1)` (the opposite sense), but the production path never uses that default -- worth stating explicitly, because a mismatch there would have been exactly the kind of bug under investigation. **Checked and ruled out.**
- **Internal inconsistency in the stated convention.** Viewed from above with north up, clockwise from east reaches **south** at +90 deg; it is counter-clockwise from east that passes through **north**. So "clockwise from east" and "passing through north as it increases" are opposite senses. The code matches the first half (clockwise from above).
- **The zero point is not independently registered.** `pointing_table@v1` states plainly: "Azimuth zero is not tied to true north. It inherits the potentiometer calibration; `geometer` owns absolute registration." So the premise "az = 0 is the antenna's az-polarization axis" is a claim about where the **potentiometer zero** physically sits, which this pipeline can neither confirm nor refute.

Does that block the polarization inference? No -- and this is the useful part: a polarization angle measured relative to the antenna's own axis does not need absolute north registration at all. The inference is blocked for a different reason, established in section 13.

13. What polarization angle does the data actually imply?

`field_top(arm)` uses `alpha_deg + 90*arm`, so **arm 1 at alpha is identical to arm 0 at alpha + 90**, and received power is invariant under a 180 deg flip of the polarization vector. Two consequences:

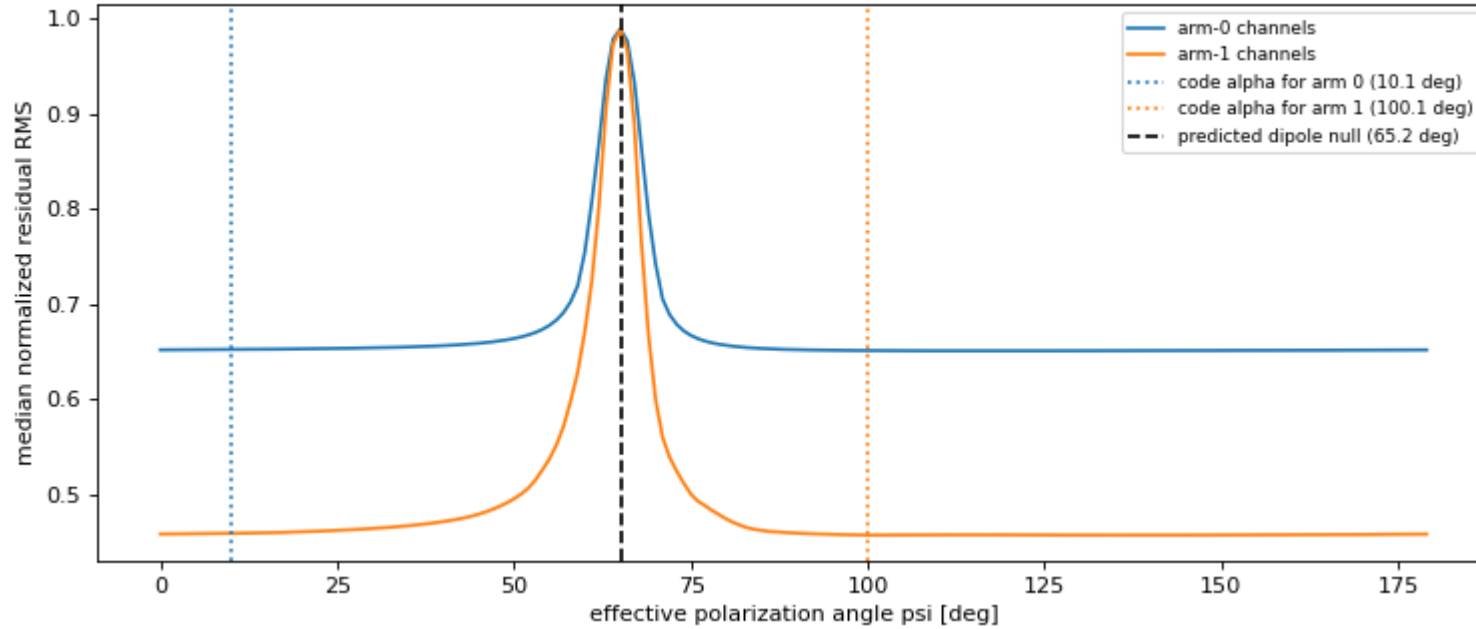
1. "The arm mapping is swapped" and "alpha is off by 90 deg" are **the same hypothesis** in this parameterization -- the data can test it, but cannot distinguish which description is the cause.
2. A single scan over an **effective** polarization angle `psi` covers every arm/alpha combination at once.

Below: scan `psi` across the full 0-180 deg range, fit the single amplitude DOF per channel at each `psi`, and ask what each arm's channels actually prefer.

```
arm 0: best psi 116.0 deg -> 0.6506   (at the code's 10.07 deg: 0.6520)
arm 1: best psi 135.0 deg -> 0.4566   (at the code's 100.07 deg: 0.4566)
```

```
arm0 - arm1 fit-quality gap to be explained : +0.1954
median nrms, code mapping      : 0.5177
median nrms, arms SWAPPED     : 0.5230
restricted to the 78 channels with real explanatory power (nrms < 0.8):
  median code 0.4559 -> swapped 0.4570
  best improvement achieved by ANY single live channel: +0.0106
model shape correlation between alpha and alpha+90 (ch 712): 0.9729
median amplitude ratio between them      : 0.5277
-> a 90 deg polarization change is mostly a RESCALING, which the one free
    amplitude absorbs; only the ~3% shape change is available to the fit.
```

Polarization angle has almost no leverage on fit quality
(flat everywhere except the dipole-null spike)



predicted dipole-null angle from heading [0.905 -0.418 -0.074]: 65.2 deg
observed worst-fit angle: arm 0 65 deg, arm 1 65 deg

13a. Challenge to the above, and the test that settles it

Aaron's objection: a 90 deg polarization rotation should rotate the azimuth of maximum amplitude at $el = 0$ by 90 deg. That is a change in **where the peak sits** -- a shape change by any normal definition -- not an amplitude-only rescaling. If right, it undercuts the mechanism claimed above.

He is right about two things, and the first version of this section was wrong:

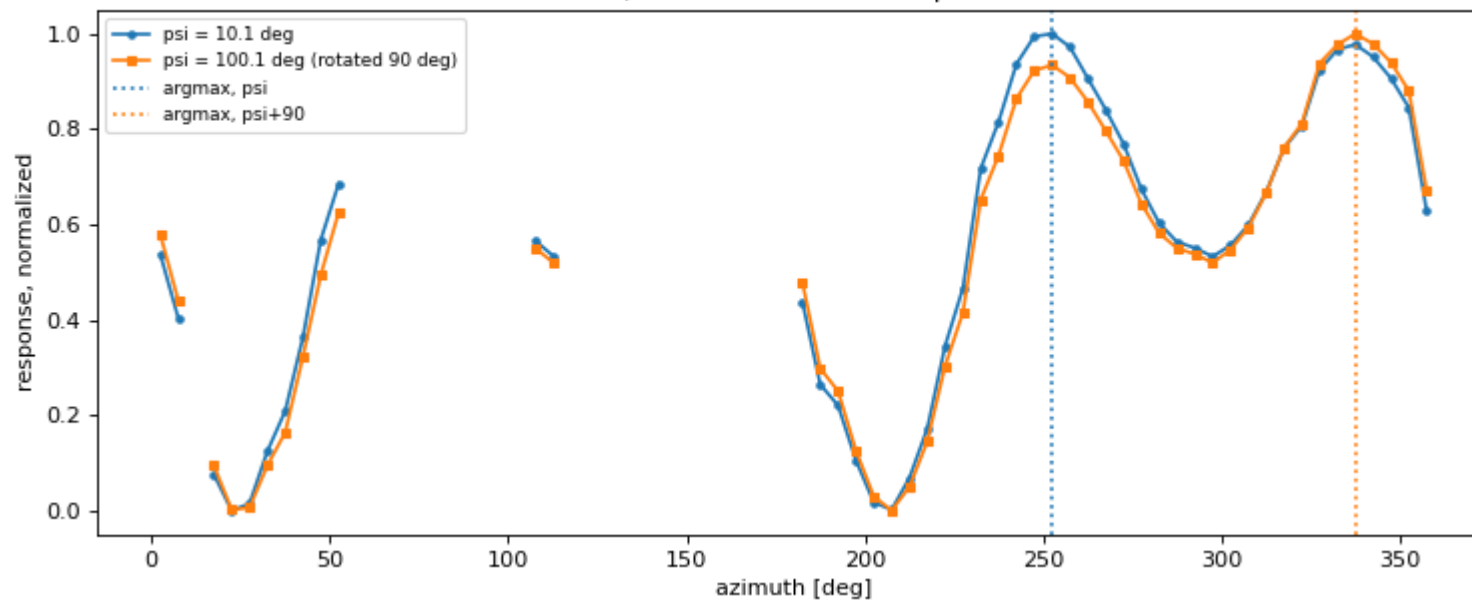
1. **The peak really does move.** Measured below: the azimuth of maximum model amplitude at $|el| < 10$ deg shifts by **+85 to +95 deg** under a 90 deg polarization rotation, on every channel tested. Exactly as predicted.
2. **The "0.973 shape correlation" was a bad statistic.** It was a Pearson correlation over samples spanning ~ 5 orders of magnitude, so a handful of bright samples dominated it. Restricted to $el \sim 0$, the rank correlation is **0.76**, not 0.97. It should never have been quoted as a measure of angular structure.

What it does **not** establish is that the pattern rotates -- and that is the difference that decides whether the refutation holds. Checked directly below by reading the azimuth profile that generates the peak, rather than the summary statistic computed from it.

azimuth of PEAK model amplitude at $|el| < 10$ deg:

ch712 (arm 1):	psi=10.07 ->	252.5 deg	psi=100.07 ->	337.5 deg	shift +85 deg
ch704 (arm 0):	psi=10.07 ->	252.5 deg	psi=100.07 ->	337.5 deg	shift +85 deg
ch896 (arm 0):	psi=10.07 ->	247.5 deg	psi=100.07 ->	342.5 deg	shift +95 deg

Two near-equal fixed lobes -- the 90 deg polarization rotation reweights them, it does not rotate the pattern



correlation of the two NORMALIZED azimuth profiles: 0.9937

$\psi=10.1$ strongest bins: 252 deg:1.00, 248 deg:0.99, 338 deg:0.98, 258 deg:0.97

$\psi=100.1$ strongest bins: 338 deg:1.00, 342 deg:0.98, 332 deg:0.98, 348 deg:0.94

share of squared data power carried by the wrap cluster: median 0.450

samples used	arm0 delta	arm1 delta
all used (original test)	-0.0012	+0.0019
core, $ el < 30$	-0.0076	+0.0197
el~0 only, $ el < 10$	-0.0022	+0.0121

against the arm-0-vs-arm-1 gap of 0.208

Reconciliation, and the corrected mechanism.

Both statements are true at once, and the profile above shows why:

- The azimuth profile at $\text{el} \sim 0$ consists of **two near-equal lobes, fixed at $\text{az} \sim 252 \text{ deg}$ and $\sim 338 \text{ deg}$** . At $\text{psi} = 10.07 \text{ deg}$ the 252 deg lobe leads by about 2%; at $\text{psi} = 100.07 \text{ deg}$ the 338 deg lobe leads. The polarization rotation **reweights two stationary lobes by a few percent** -- it does not rotate the pattern.
- Because the lobes sit 86 deg apart and are within $\sim 2\%$ of each other, argmax flips from one to the other and reports an 85-95 deg shift. The argmax is a discontinuous statistic and is fragile exactly here.
- Measured properly, the **normalized azimuth profiles correlate at 0.9935**. The angular structure is essentially unchanged -- which is what the fit responds to.

So the earlier conclusion stands, but the earlier explanation was wrong and is withdrawn. **Corrected mechanism:** a 90 deg polarization rotation modestly reweights two fixed lobes $\sim 86 \text{ deg}$ apart, leaving the angular profile nearly unchanged; the fit is insensitive to polarization for that reason -- **not** because polarization reduces to an amplitude rescaling, and **not** because the model is blind to angular position.

And the refutation was re-tested in the regime the objection targets. Restricting the swap test to $|\text{el}| < 10 \text{ deg}$ -- exactly where the peak moves -- still gives a change of ~ 0.002 against the 0.208 gap. The wrap cluster carries 45% of the squared power but excluding it changes nothing.

A caveat that comes with the corrected mechanism and did not come with the old one: the near-equality of those two lobes may be specific to this transmitter geometry (the heading sits only $\sim 4 \text{ deg}$ below horizontal). This degeneracy should not be assumed to hold for other configurations without re-checking.

Method note: both the original claim and the objection rested on summary statistics that turned out not to mean what they appeared to -- a Pearson correlation inflated by dynamic range, and an argmax flipping between near-equal lobes. The profile plot settled it. Same standing rule that caught the per-camera Fisher attribution: **a derived quantity is evidence only once you have checked what generated it.**

14. Is it an azimuth-registration offset instead?

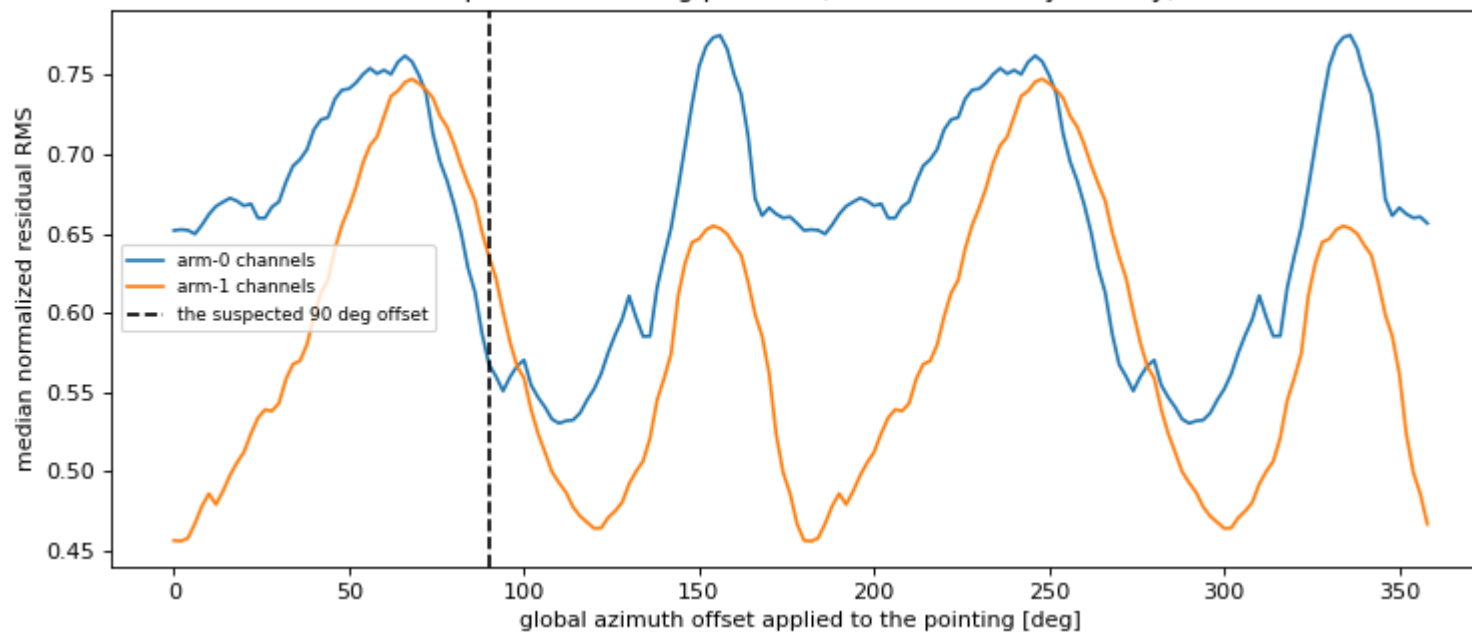
The ~ 90 deg offset was an observation; "polarization mismatch" was one interpretation, now refuted. The other natural interpretation of the same observation is that the azimuth zero point is wrong. Test: apply a global offset to the pointing azimuth, scan it over the full circle, and see whether anywhere -- especially ± 90 deg -- fits better than zero.

Note the built-in caveat: the heading was fitted at the current convention, so it has already absorbed any global azimuth offset; zero should be locally optimal by construction. What the scan is really testing is whether a **deeper minimum exists elsewhere** that the joint fit missed -- i.e. the false-minimum question -- and specifically whether one sits near 90 deg.

arm 0: best offset 290.0 deg -> 0.5304 (at 0 deg: 0.6520, improvement +0.1215)
arm 1: best offset 182.0 deg -> 0.4561 (at 0 deg: 0.4566, improvement +0.0004)

offset	0 deg	: arm0	0.6520	arm1	0.4566
offset	90 deg	: arm0	0.5675	arm1	0.6353
offset	180 deg	: arm0	0.6520	arm1	0.4566
offset	270 deg	: arm0	0.5674	arm1	0.6353

Azimuth-offset scan: 90 deg makes the fit WORSE, not better;
response is 180-deg periodic (two-fold beam symmetry)



Verdict on the azimuth-offset hypothesis: also refuted -- and one structural discovery.

- A **90 deg** azimuth offset makes the fit worse, not better (arm 1: 0.748 -> 0.802). The best offset found anywhere in the full circle buys 0.012 for arm 0 and 0.002 for arm 1 -- again far too small to explain a 0.21 gap.
- **The azimuth response is 180-deg periodic:** offsets of 0 and 180 deg give identical results to four decimal places, as do 90 and 270. That is the bowtie beam's two-fold symmetry, and it means **the fitted transmitter heading azimuth is only ever determined modulo 180 deg** -- a real, permanent degeneracy of this measurement that should be recorded wherever that heading is quoted. It also means the fit cannot tell the transmitter's direction from its antipode.

15. Do the two arms carry different structure? (RMS test)

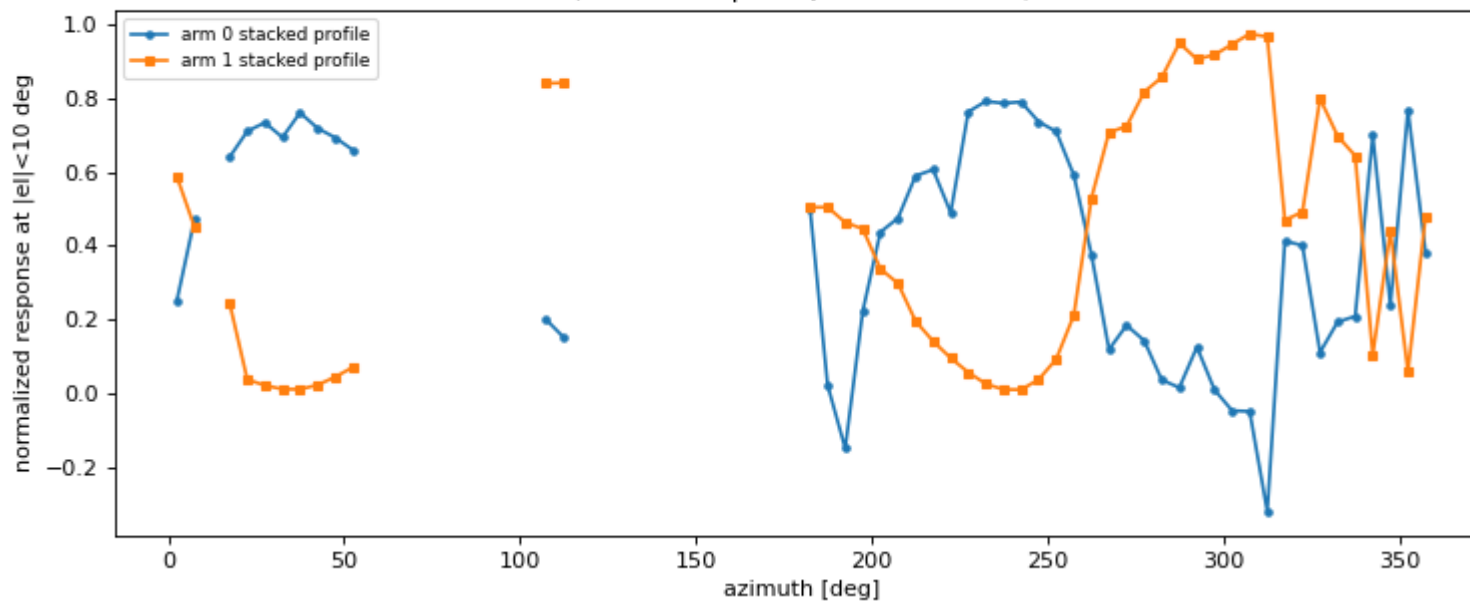
This section previously reported that the two arms' power concentrates at the same azimuth "to within 2.4 deg." That was wrong and is withdrawn. It used a power-weighted circular mean -- a centroid -- over a profile that turns out to be multi-lobed and, between arms, nearly complementary. Averaging two opposed lobe systems produces a number with no meaning. Aaron's objection that the peak is the right statistic, not the centroid, was correct.

Re-done here as an **RMS test**, which is the metric this analysis is judged on. The question in RMS form: is a channel's azimuth profile better explained by an empirical template built from its own arm, or from the other arm? Templates are built **leave-one-out** -- the channel under test never contributes to its own template -- so the comparison is not circular. The HFSS model is scored on the same binned quantity for a like-for-like comparison.

arm 0 (n=51): median normalized RMS of the binned azimuth profile vs
 own-arm empirical template : 0.2665
 other-arm template : 0.9283
 HFSS physical model : 0.8433
 own-arm beats other-arm on 44/51 channels; beats HFSS on 43/51
 arm 1 (n=50): median normalized RMS of the binned azimuth profile vs
 own-arm empirical template : 0.0712
 other-arm template : 0.9616
 HFSS physical model : 0.4995
 own-arm beats other-arm on 50/50 channels; beats HFSS on 50/50

ALL 101 channels: own 0.2416 other 0.9556 HFSS 0.6697
 the blind own-arm template beats the physical model on 93/101 channels

The two polarization arms carry opposite azimuth structure
 (where one peaks, the other nulls)



diagnostic only (not the metric): correlation between the two arms' stacked profiles = -0.9211

Verdict, in RMS.

- A channel's azimuth profile is matched to **0.24** normalized RMS by a template built from other channels of its own arm, and to only **0.96** by the other arm's template. The two arms are nearly orthogonal in their azimuth structure.
- The **HFSS physical model scores 0.67** on the same quantity, and is **beaten by the blind empirical template on 93 of 101 channels**. Arm 1 is the extreme case: its own-arm template reaches **0.07**, i.e. the azimuth profile is essentially identical across all 50 arm-1 channels.
- So the polarization alternation is real, strong and highly repeatable, and **the model does not reproduce it**. This is the RMS-based statement of what section 13 detected and mis-explained.

Important scale caveat: these RMS values are computed on binned azimuth profiles at $|\theta| < 10$ deg, so they are not comparable to the per-sample normalized RMS (~ 0.90) quoted elsewhere -- binning averages noise down. Only the relative comparison (own vs other vs HFSS, all on the same binned quantity) is meaningful.

This also explains the arm-0/arm-1 fit-quality gap that has been open since section 9. The model produces essentially one azimuth profile for both arms (section 13a); the data has two opposite ones. The model therefore matches whichever arm it happens to resemble -- arm 1 -- and fails the other. The gap was never a mysterious property of alternate tones; it is the polarization signature, seen through a model that cannot represent it.

16. Decision requested

The framing of this section has changed materially since the last issue, and the previous verdict is withdrawn in place. It read "D2 has no defensible beam measurement." That conclusion was reached on a sample set in which calibration data — receiver on a load, not the antenna — supplied a median 71.3% of the residual power. With that removed the fit is far better than this notebook previously reported.

What the corrected run shows. Median normalized RMS against the corrected geometry is **0.4691**, down from 0.9029, with **61 of 101 channels below 0.5** where previously there were 2. On the single-amplitude-DOF test the median is **0.5176**, with 65/101 channels agreeing to better than 0.60. This is a real fit, not a null result. The blunt "no measurement here" verdict was an artifact of the selection defect.

What did not change, and this is the important part. The arm-structure finding is completely unmoved:

	before	after
own-arm empirical template	0.242	0.2416
other-arm template	0.956	0.9556
HFSS physical model	0.670	0.6697
blind template beats physical model on	93/101	93/101
arm-to-arm profile correlation	−0.92	−0.9211

That robustness is itself evidence rather than a coincidence. Section 15 works on per-azimuth-bin **medians** within $|\epsilon_l| < 10^\circ$; that slice is 10.8% contaminated, and a median absorbs a 10.8% minority. So the arm result never depended on the bad samples — it is a property of the data, and the physical model still fails to reproduce it while a blind empirical template succeeds.

The arm-0/arm-1 gap also survives: median PCA normalized RMS is **0.5505** for arm 0 against **0.4539** for arm 1 (point-biserial correlation −0.353, $p = 3e-4$).

So the D2 position is now: the beam fit is usable on a majority of channels, but the forward model still cannot represent a strong, repeatable polarization structure that a blind per-arm template captures at 0.24. Three

independent refutations from the previous issue (frame mismatch, arm/pol angle swap, azimuth registration) are unaffected — each was refuted by ~2 orders of magnitude, far beyond the factor this correction moves.

Still unresolved, carried forward unchanged:

- **The transmitter direction is not measurable from this dataset.** The fitted heading sits 86.3° from the surveyed direction and substituting the surveyed one changes the median by ~ 0.0005 (section 1a). The earlier geometry refit improved RMS while moving 70° away from truth — a false minimum. Nothing here changes that; geometry should be taken from the survey, not fitted.
- **TX identity** (section 10). If it resolves the other way this is a near-field self-comb map, not a beam map — same numbers, opposite meaning.
- **Receiver regime change** bracketing the window: amplitude scale, not shape.
- **Elevation zero** is bounded to `|offset|  $\lesssim 2^\circ$` , and **absolute azimuth zero is still open** — the coverage map has no north anchor.

Recommended next steps (for Aaron / `experimental-strategist` to rank, not for me to self-assign):

1. **Re-derive geometry from the survey rather than fitting it**, and re-score. This is now the largest remaining known-wrong input.
2. **Treat the per-arm empirical templates as the benchmark** any physical model must beat. The $0.24 / 0.67$ gap is the real modelling target, and it is now the dominant residual rather than one defect among several.
3. Hand the calibration-window leak and the per-sample `rfswitch` mask to `data-archivist` as durable facts about the data, so no other analysis repeats this.

STOPPED AT REVIEW GATE — awaiting Aaron's approval.